1) a) $f(x) = x^3 \cdot e^{1+2x}$ $\Rightarrow$ Produktregel $\quad u \cdot v \Rightarrow u' \cdot v + v' \cdot u$

$u = x^3 \qquad u' = 3x^2$

$v = e^{1+2x} \qquad v' = 2 \cdot e^{1+2x}$

$$f'(x) = 3x^2 \cdot e^{1+2x} + 2e^{1+2x} \cdot x^3$$
$$= e^{1+2x} \cdot (2x^3 + 3x^2)$$

b) $f(x) = \dfrac{2x}{2x^2+1}$ $\Rightarrow$ Quotientenregel $\quad \dfrac{u}{v} \Rightarrow \dfrac{u' \cdot v - v' \cdot u}{v^2}$

$u = 2x \qquad u' = 2$

$v = 2x^2+1 \qquad v' = 4x$

$$f'(x) = \frac{2 \cdot (2x^2+1) - 4x \cdot 2x}{(2x^2+1)^2}$$
$$= \frac{-4x^2+2}{(2x^2+1)^2}$$

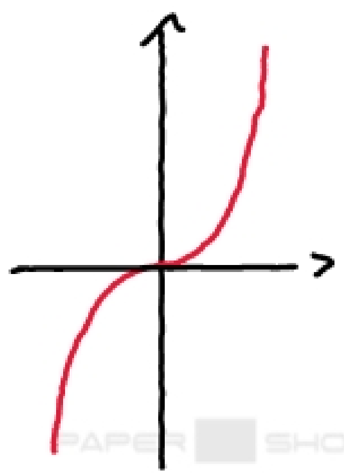
c) $f(x) = (x^3 + \sqrt{3}) \cdot \ln(x^2+x)$

$u = x^3 + \sqrt{3} \qquad u' = 3x^2$

$v = \ln(x^2+x) \qquad v' = \dfrac{2x+1}{x^2+x}$

$$f'(x) = 3x^2 \cdot \ln(x^2+x) + (x^3+\sqrt{3}) \cdot \frac{2x+1}{x^2+x}$$

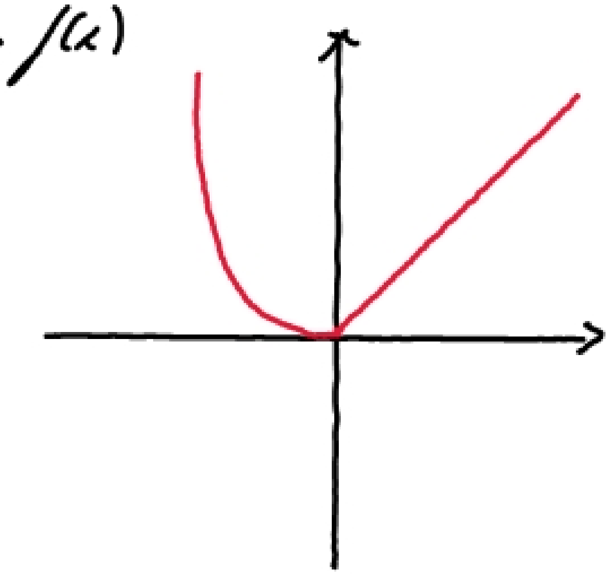
d) $f(x) = x \cdot |x| = \begin{cases} x^2 & ; x \geq 0 \\ -x^2 & ; x < 0 \end{cases}$ $\Rightarrow$ $f'(x) = \begin{cases} 2x & ; x \geq 0 \\ -2x & ; x \leq 0 \end{cases}$



$\underbrace{\lim_{x \to 0^+} f(x) = f(0) = \lim_{x \to 0^-} f(x)}_{\text{stetig}} \wedge \underbrace{\lim_{x \to 0^+} f'(x) = f'(0) = \lim_{x \to 0^-} f(x)}_{\text{differenzierbar}}$

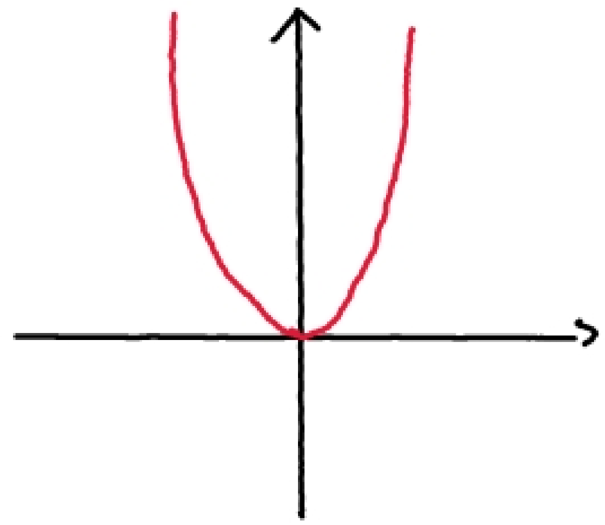
2) a) $f(x) = \begin{cases} x \, ; & x \geq 0 \\ x^2 \, ; & x < 0 \end{cases}$ stetig: $\lim\limits_{x \to 0^+} f(x) = f(0) = \lim\limits_{x \to 0^-} f(x)$

$$\left.\begin{aligned} &\lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0^+} \frac{x - 0}{x - 0} = 1 \\ &\lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0^-} \frac{x^2 - 0}{x - 0} = \lim_{x \to 0^-} x^2/x = 0 \end{aligned}\right\} \text{stetig aber nicht differenzierbar}$$



b) $f(x) = \begin{cases} x^3 & ; x \geq 0 \\ x^2 & ; x < 0 \end{cases}$ stetig: $\lim\limits_{x \to 0^+} f(x) = f(0) = \lim\limits_{x \to 0^-} f(x)$

$$\left.\begin{aligned} &\lim_{x \to 0^+} \frac{f(x) - 0}{x - 0} = \lim_{x \to 0^+} \frac{x^3 - 0}{x - 0} = \lim_{x \to 0^+} x^3/x = 0 \\ &\lim_{x \to 0^-} \frac{f(x) - 0}{x - 0} = \lim_{x \to 0^-} \frac{x^2 - 0}{x - 0} = \lim_{x \to 0^-} x^2/x = 0 \end{aligned}\right\} \text{stetig und auch differenzierbar}$$



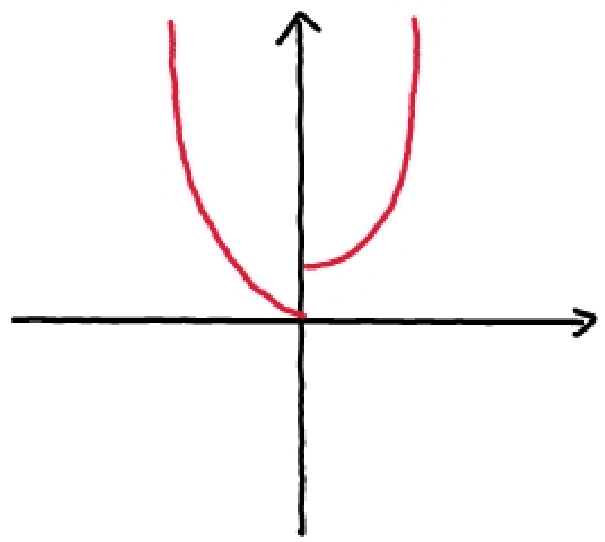
c) $f(x) = \begin{cases} x^3 + 1 & ; x \geq 0 \\ x^2 & ; x < 0 \end{cases}$

$\lim\limits_{x \to 0^+} f(x) = f(0) = 1$

$\lim\limits_{x \to 0^-} f(x) = 0$

$\Downarrow$

nicht stetig und somit auch nicht differenzierbar



3) $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$ $\quad f(x) = x^2 \rightarrow f'(x) = 2x$

$f^{-1}(x) = \sqrt{x}$

$$\Rightarrow (\sqrt{x})' = \frac{1}{2 \cdot f^{-1}(x)} = 1/2\sqrt{x}$$

4) $f(x) = \sum_{k=0}^{\infty} x^k \rightarrow f'(x) = \sum_{k=1}^{\infty} k \cdot x^{k-1} = \sum_{k=0}^{\infty} (k+1) \cdot x^k$ $\quad x \in (-1;1)$

$g(x) = \frac{1}{1-x} = (1-x)^{-1} \rightarrow g'(x) = (-1) \cdot (1-x)^{-2} \cdot (-1) = \frac{1}{(1-x)^2}$

Es gilt $f(x) = g(x)$: $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$ geometrische Reihen

somit auch $f'(x) = g'(x)$: $\sum_{k=0}^{\infty} (k+1) \cdot x^k = \frac{1}{(1-x)^2}$

5) a) $f(x) = x^a = (e^{\ln x})^a = e^{a \cdot \ln x}$

$f'(x) = e^{a \cdot \ln x} \cdot (a \cdot \ln x)' = e^{a \ln x} \cdot (a \cdot 1/x) = x^a \cdot a/x = a \cdot x^{a-1}$

b) $f(x) = c^x = (e^{\ln c})^x = e^{x \cdot \ln c}$

$f'(x) = e^{x \cdot \ln c} \cdot (x \cdot \ln c)' = e^{x \cdot \ln c} \cdot \ln c = \ln c \cdot c^x$

6) $f(x) = \tan(x) = \dfrac{\sin(x)}{\cos(x)} = \dfrac{u}{v}$

$u = \sin(x) \qquad u' = \cos(x)$

$v = \cos(x) \qquad v' = -\sin(x)$

$$f'(x) = \frac{\cos(x) \cdot \cos(x) - (-\sin(x)) \cdot \sin(x)}{\cos^2(x)}$$

$$= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{\cos^2(x)}{\cos^2(x)} + \frac{\sin^2(x)}{\cos^2(x)} = 1 + \tan^2(x)$$

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

$$(\arctan(x))' = \frac{1}{1 + \tan^2(\arctan(x))} = \frac{1}{1 + x^2}$$